

EX. NO.8**FABRIC NETWORK DEPLOYMENT AND MANAGEMENT****AIM**

To deploy, manage, monitor, and maintain a Hyperledger Fabric network by creating the network, verifying network components, deploying chaincode, monitoring Docker containers, and performing network shutdown and cleanup operations.

SOFTWARE REQUIREMENTS

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code (Optional)

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

PROCEDURE**Step 1: Navigate to the Fabric Test Network**

Navigate to the Hyperledger Fabric test network directory.

```
cd ~/fabric-dev/fabric-samples/testnetwork
```

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Step 2: Start the Fabric Network

Start the Fabric network, create the application channel, and enable Certificate Authorities.

`./network.sh up createChannel -ca`

```
vboxuser@Ubuntu:~$ cd ~/fabric-dev/fabric-samples/test-network
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -ca
Using docker and docker compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'Certificate Authorities'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
```

Step 3: Verify Running Docker Containers

Display all currently active Docker containers. `docker ps`

This command can be used to verify that the peer, orderer, and Certificate Authority containers are running.

```
channel 'mychannel' joined
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
5a06f87ae151   hyperledger/fabric-orderer:latest   "orderer"              21 seconds ago Up 20 seconds 0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp orderer.example.com
48225fe83df0   hyperledger/fabric-peer:latest      "peer node start"      21 seconds ago Up 20 seconds 0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp peer0.org2.example.com
137163849e3c   hyperledger/fabric-peer:latest      "peer node start"      21 seconds ago Up 20 seconds 0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp peer0.org1.example.com
f3a96145620a   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se..." 35 seconds ago Up 35 seconds 0.0.0.0:9054->9054/tcp, [::]:9054->9054/tcp, 7054/tcp, 0.0.0.0:10054->10054/tcp, [::]:10054->10054/tcp
```

Step 4: Display Docker Images

Display the Docker images available on the system.

`docker images`

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This allows the Fabric Docker images used by the network to be verified.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use
busybox:latest	dc2d74b28e4c	6.81MB	2.23MB	
ghcr.io/hyperledger/fabric-baseos:2.5.16	109b4a15b282	257MB	83.2MB	
ghcr.io/hyperledger/fabric-ca:1.5.17	453a0a727e82	381MB	123MB	U
ghcr.io/hyperledger/fabric-ccenv:2.5.16	807493f095f6	1.01GB	257MB	
ghcr.io/hyperledger/fabric-orderer:2.5.16	1d918fc7e1e0	182MB	50.7MB	U
ghcr.io/hyperledger/fabric-peer:2.5.16	58979b65744b	232MB	66.8MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U
hyperledger/fabric-baseos:2.5	109b4a15b282	257MB	83.2MB	
hyperledger/fabric-baseos:2.5.16	109b4a15b282	257MB	83.2MB	

Step 5: Check Network Status

Display all Docker containers, including stopped containers.

```
docker ps -a
```

This command helps monitor the current status of Fabric network containers.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED
5a06f87ae151	hyperledger/fabric-orderer:latest	"orderer"	About a minute ago
Up About a minute	0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp	orderer.example.com	
48225fe83df0	hyperledger/fabric-peer:latest	"peer node start"	About a minute ago
Up About a minute	0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp	peer0.org2.example.com	
137163849e3c	hyperledger/fabric-peer:latest	"peer node start"	About a minute ago
Up About a minute	0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp	peer0.org1.example.com	
f3a96145620a	hyperledger/fabric-ca:latest	"sh -c 'fabric-ca-se..."	About a minute ago
Up About a minute	0.0.0.0:9054->9054/tcp, [::]:9054->9054/tcp, 7054/tcp, 0.0.0.0:19054->19054/tcp, [::]:19054->19054/tcp	ca_orderer	
4f79edcac9b1	hyperledger/fabric-ca:latest	"sh -c 'fabric-ca-se..."	About a minute ago
Up About a minute	0.0.0.0:7054->7054/tcp, [::]:7054->7054/tcp, 0.0.0.0:17054->17054/tcp, [::]:17054->17054/tcp	ca_org1	
1706553f01c2	hyperledger/fabric-ca:latest	"sh -c 'fabric-ca-se..."	About a minute ago
Up About a minute	0.0.0.0:8054->8054/tcp, [::]:8054->8054/tcp, 7054/tcp, 0.0.0.0:18054->18054/tcp, [::]:18054->18054/tcp	ca_org2	

Step 6: Deploy Asset Transfer Chaincode

Deploy the JavaScript Asset Transfer chaincode to the Fabric network.

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transferbasic/chaincode-javascript
```

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```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl javascript
t -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Using docker and docker compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
```

Step 7: Verify Installed Chaincode

Query the installed chaincode

packages. peer lifecycle

chaincode queryinstalled

This command verifies that the chaincode package has been installed on the peer.

Step 8: Monitor Peer Logs

Display the logs generated by the Org1

peer node. docker logs

peer0.org1.example.com

The logs can be used to monitor peer startup, connections, transactions, and chaincode-related activities.

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```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ source setOrg1.sh
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer lifecycle chaincode queryinstall
led
Installed chaincodes on peer:
Package ID: basic_1.0:1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629, Label:
basic_1.0
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker logs peer0.org1.example.com
2026-08-31 16:32:45.136 UTC 0001 INFO [nodeCmd] serve -> Starting peer:
Version: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64
Chaincode:
Base Docker Label: org.hyperledger.fabric
Docker Namespace: hyperledger
2026-08-31 16:32:45.141 UTC 0002 INFO [nodeCmd] serve -> Peer config with combined core.yaml settings and environment variable overrides:
chaincode:
  builder: $(DOCKER_NS)/fabric-ccenv:${TWO_DIGIT_VERSION}
  executetimeout: 300s
  externalbuilders:
```

Step 9: Monitor Orderer Logs

Display the logs generated by the orderer node.

`docker logs orderer.example.com`

The logs can be used to monitor the ordering service and transaction processing.

```
hyperledger-fabric@v2.5.16$ docker logs orderer.example.com
2026-08-31 16:32:45.992 UTC 0001 INFO [localconfig] completeInitialization -> Kafka.Version un
set, setting to 0.10.2.0
2026-08-31 16:32:45.993 UTC 0002 INFO [orderer.common.server] prettyPrintStruct -> Orderer con
fig values:
  General.ListenAddress = "0.0.0.0"
  General.ListenPort = 7050
  General.TLS.Enabled = true
  General.TLS.PrivateKey = "/var/hyperledger/orderer/tls/server.key"
  General.TLS.Certificate = "/var/hyperledger/orderer/tls/server.crt"
  General.TLS.RootCAs = [/var/hyperledger/orderer/tls/ca.crt]
  General.TLS.ClientAuthRequired = false
  General.TLS.ClientRootCAs = []
  General.TLS.TLSHandshakeTimeShift = 0s
  General.Cluster.ListenAddress = ""
  General.Cluster.ListenPort = 0
  General.Cluster.ServerCertificate = ""
  General.Cluster.ServerPrivateKey = ""
  General.Cluster.ClientCertificate = "/var/hyperledger/orderer/tls/server.crt"
  General.Cluster.ClientPrivateKey = "/var/hyperledger/orderer/tls/server.key"
  General.Cluster.RootCAs = [/var/hyperledger/orderer/tls/ca.crt]
```

Step 10: Verify the Channel List

the channels joined by the

peer. `peer channel list`

This verifies that mychannel has been successfully created and joined by the peer.

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Step 11: Stop the Fabric Network

Stop and remove the Fabric network.

`./network.sh down`

This removes the Fabric containers, network, and generated network artifacts.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
2026-08-31 16:39:48.792 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Channels peers has joined:
mychannel
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker compose
Stopping network
[+] down 3/6
  * Container ca_org1          Stopping          2.0s
  ✓ Container peer0.org1.example.com Removed          1.7s
  ✓ Container peer0.org2.example.com Removed          1.8s
  * Container ca_org2          Stopping          2.0s
  * Container ca_orderer       Stopping          2.0s
  ✓ Container orderer.example.com Removed          1.2s
```

Step 12: Clean Docker Resources

Optionally remove unused Docker containers, networks, images, and other resources. `docker system prune -f`

This command helps reclaim unused Docker resources and disk space.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker system prune -f
Deleted Containers:
4a6cedf6a948d1759e031f6e82e945acc0df6eb44d6522398904d9c911b0966f
5fff00374402057deb1eb301f9862b2bfe4ddff6abfd97e90fd35f1b9e0ded9e
fab10d165a35b7b84e12a63267468588994df14c99c9969b122b3cd53e2e9715
c848a85aa9cdf9e0bffc7213eb83bc67e99c68af52eb501f065951f39c2c4e4
74a0cdba564892b8daf6f98b08439cd60fe670e294c2e74b5e16c2e8827af82c
21a81efbebe635cd4d0c7fbbd7aefc65aa521664353d628702b76c90c663cf9d
2c4df2621ee55d9d1e6d0eac8051d5d67e33fc4c632ee59e083b69a4ea08905
022bb13a6257d24c24d464eae8d52a2fb1f4e9ec501bc0fa0f68ab07e3094240
71e6a51cd89978e9b72aa305d72b41e08fd049a4a4d479d5def250828afa3767
f4cfe3a8fc8c98d438b03253792785e60ec8d26f5f231d923cac86365c0daf8d
371e53b7ed5ced922b84068eb8db73eadfb6d14557acead9f066477be769952e
ae6daf8406e38cb66e273fd7bb1b843ea24035174b908706aab43916ad395a83
5d3201051db4744924f566254e23388709a856855a4d6291589e8ef5e442723e
5d3201051db4744924f566254e23388709a856855a4d6291589e8ef5e442723e
```

NETWORK MANAGEMENT WORKFLOW

Install Dependencies

|



Start Fabric Network

|



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Create Channel

|



Verify Docker Containers

|



Deploy Chaincode

|



Verify Installed Chaincode

|



Monitor Peer Logs

|



Monitor Orderer Logs

|



Verify Channel

|



Stop Network

|



Clean Docker Resources

RESULT

The Hyperledger Fabric network was successfully deployed and managed. The peer nodes, ordering service, Certificate Authorities, Docker containers, and application channel were verified. JavaScript chaincode was deployed and

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its installation was verified. Peer and orderer logs were monitored, and the network was successfully stopped and cleaned up.